

VELMÈRE BASIC AUDIT REPORT

Tether USD (0xdac17f958d2ee523a2206206994597c13d831ec7)

VELMERE SECURITY AUDIT REPORT: TETHER USD

Network: Ethereum Mainnet (Chain ID: 1)
Report ID: vlm_usdt_basic_1 | Tier: BASIC | Application Surface: BROWSER
Created At: 2026-09-07T21:37:57.394Z

VERDICT SUMMARY: MODERATE RISK (42/100)

Confidence Score: 95/100 | Evidence Coverage: 96%
Tether employs a centralized administrative governance model with unilateral address blacklisting (addBlackList) and token destruction capabilities without timelock.

--- AUDIT OVERVIEW & CONTRACT CONTEXT [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Summary of execution scope and target objectives
Project / Contract: Tether USD
Contract Address: 0xdac17f958d2ee523a2206206994597c13d831ec7
Network / Chain ID: Ethereum Mainnet (ID: 1)
Token / Symbol: USDT
Proxy Pattern: Upgradeable via Custom Upgrade Proxy
Compiler Version: solc 0.4.18

--- SOURCE CODE VERIFICATION & BYTECODE DECOMPILATION [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
State hash match against blockchain node and dispatcher analysis
EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

--- BASELINE SECURITY FINDINGS [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Identified vulnerabilities in baseline scan

Findings:

- [MEDIUM] VLM-USDT-01: Legacy Solidity Compiler (v0.4.18)
Category: Legacy Compiler | State: recommended
Evidence: pragma solidity ^0.4.17; in TetherToken.sol
Recommendation: Ensure off-chain integration wrappers protect against unhandled revert bubbles.
- [LOW] VLM-USDT-02: Non-standard ERC20 Return Values
Category: ERC20 Conformance | State: recommended
Evidence: function transfer(address _to, uint _value) public; (missing returns (bool))
Recommendation: Always interact via OpenZeppelin SafeERC20 safeTransfer() primitives.

--- PERMISSION & ROLE GOVERNANCE MAP (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Deep analytical map of privileged roles, owner powers, and timelock bounds
[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Admin role: privilege and access control hierarchy in bytecode
 - * Verification of timelock delay parameters and governance controls
 - * Detection of privileged mint, pause, blacklist, and drainage capabilities
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

VELMÈRE BASIC AUDIT REPORT

Tether USD (0xdac17f958d2ee523a2206206994597c13d831ec7)

--- LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Token distribution curve, whale clusters, and verifiable DEX lock proofs

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Token distribution curve and top holder concentration analysis
 - * DEX liquidity pool pairing and lock verification across registered vaults
 - * Automated honeypot simulation, transfer fee checks, and sell restriction audit
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Simulation of oracle manipulation, flash loans, and cross-function reentrancy

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Cross-function and cross-contract reentrancy guard modeling
 - * Price oracle staleness, manipulation resilience, and market feed verification
 - * Economic shock modeling and flash-loan manipulation resistance simulation
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Deep storage layout slot collision detection across implementation upgrades

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Storage slot allocation collision detection across implementation upgrades
 - * Upgrade proxy layout gap verification and state preservation checks
 - * Bytecode reproduction against official compiler metadata and build pipelines
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- HISTORICAL VERSION EVOLUTION & REGRESSION PROOFS (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Line-by-line semantic diff of logic changes between production contract generations

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Version evolution and business logic change tracking
 - * Bytecode regression analysis of active implementation
 - * Formal verification: NOT EXECUTED (Formal verification engine not commissioned)
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED) [ADVANCED] ---

VERIFICATION TYPE: INDEPENDENT MANUAL AUDITOR REVIEW

VELMÈRE BASIC AUDIT REPORT

Tether USD (0xdac17f958d2ee523a2206206994597c13d831ec7)

Attested reviewer evidence and signed non-automated verification findings
[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Analyst sign-off requires verified reviewer evidence
 - * Cryptographically signed attestation digest
 - * Manual review boundary explicit: no certified safe claims
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

CONFIDENTIALITY & LEGAL NOTICE:

This document is an evidence-bound security analysis report produced by Velmere Security.
It does not represent a commercial warranty, investment advice, or guaranteed-safe certification.